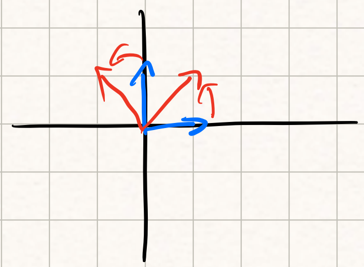
Last time:

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$



→ causes rotation!

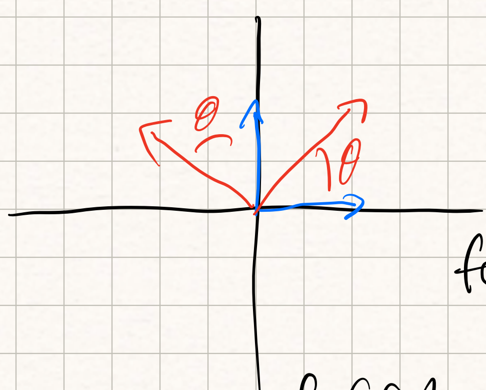
$$\det\begin{bmatrix} 1-\lambda & -1 \\ 1 & 1-\lambda \end{bmatrix} = 0 = (1-\lambda)^2+1 = \lambda^2-2\lambda+2$$

$$\lambda = \frac{2 \pm \sqrt{4-4(2)}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$$

$$\begin{bmatrix} 1-(1-i) & -1 \\ 1 & 1-(1-i) \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 0 \qquad ix - y = 0 \qquad y = ix \qquad v_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$\begin{bmatrix} 1-(1+i) & -1 \\ 1 & 1-(1+i) \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 0 \qquad -ix = y \qquad v_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

rigid rotation around $(0,0)$ in $\mathbb{R}^2$ is a linear transformation, what is the matrix for it



$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

for rotation by $\frac{\pi}{4}$ $\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$

for rot. by $\frac{3\pi}{2}$ $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

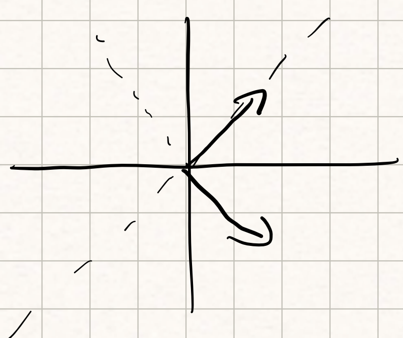
rotation are linear and have complex eigenvalues

dilation is linear and have real eigenvalues

ex dilation by 7 $\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$ $\lambda_1, \lambda_2 > 0$

reflection over a line is linear and has real eigenvalues

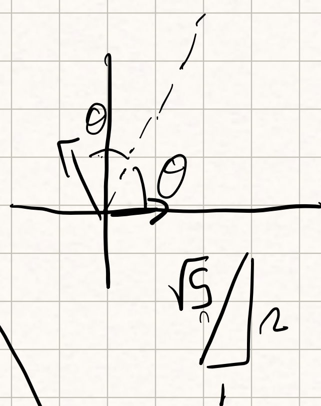
$\lambda_1 < 0$



any linear transformation $\mathbb{R}^2 \to \mathbb{R}^2$ is a composition of reflection, dilation and rotation

Write the matrix for

1) dilate by $\frac{1}{2}$ $\begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$
2) reflect over $y = 2x$ $\begin{bmatrix} \; & \; \\ \; & \; \end{bmatrix}$
3) rotate by $\frac{\pi}{3}$ $\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$



$$\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} \; & \; \\ \; & \; \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

in eigen vector basis $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

basis v's $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

reflection in eigenbasis

$$\underset{\text{standard basis}}{\mathbb{R}^2} \xrightarrow{\Phi} \underset{\text{eigenvalue basis}}{\mathbb{R}^2} \xrightarrow{A} \mathbb{R}^2 \xrightarrow{\Phi^{-1}} \mathbb{R}^2$$

$$\Phi^{-1} A \Phi$$

$$\begin{bmatrix} & \\ & \end{bmatrix}\begin{bmatrix} 1 & 6 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & -1 & 0 & 1 \end{array}\right] \quad \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -3 & -2 & 1 \end{array}\right] \quad \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 2/3 & -1/3 \end{array}\right] \quad \left[\begin{array}{cc|cc} 1 & 0 & -1/3 & 2/3 \\ 0 & 1 & 2/3 & -1/3 \end{array}\right]$$

$$\Phi^{-1} = \begin{bmatrix} -1/3 & -2/3 \\ 2/3 & -1/3 \end{bmatrix}$$

$$\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}\begin{bmatrix} -1/3 & -2/3 \\ 2/3 & -1/3 \end{bmatrix}\begin{bmatrix} 1 & 6 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$